

Khadija Amber

Khadijaamber44@gmail.com | +92 300 9754077 | linkedin.com/in/Khadija-Amber | github.com/Khadija-Amber

PROFESSIONAL SUMMARY

Computer Science undergraduate (CGPA: 3.4/4.0) with hands-on experience in full-stack web development, backend engineering, and AI-powered systems. Proficient in Python, Java, Django, SQL, REST APIs, database design, and software architecture. Experienced in developing scalable, secure, and high-performance applications while integrating machine learning solutions into production-oriented systems. Passionate about building efficient software solutions and leveraging modern technologies to solve real-world challenges.

TECHNICAL SKILLS

Languages: Python, C++, SQL, JavaScript, HTML, CSS
Machine Learning & AI: Machine Learning, Feature Engineering, Data Preprocessing, Exploratory Data Analysis (EDA), Model Training & Evaluation, Scikit-learn, TensorFlow, Hyperparameter Tuning, Model Selection, Cross-Validation, Selenium, AI Automation
Web Development: Django, Django REST Framework (DRF), REST APIs, API Integration, Bootstrap
Software Engineering: Object-Oriented Programming (OOP), Data Structures & Algorithms (DSA), Database Design, Agile Development
Developer Tools & Platforms: Git, GitHub, Jupyter Notebook, VS Code, PyCharm
Databases: MySQL, SQLite

EDUCATION

BS in Computer Science — Bahauddin Zakariya University, Multan, Pakistan 2023 – Present

WORK EXPERIENCE

- Academic Project Developer** | Bahauddin Zakariya University Jun 2026 – Present
- Developed full-stack web applications using Python, Django, Django REST Framework, MySQL, and REST APIs while implementing secure authentication and scalable database design.
 - Built and evaluated machine learning solutions through data preprocessing, exploratory data analysis (EDA), feature engineering, model training, hyperparameter tuning, and cross-validation.
- Undergraduate Research Assistant** | Bahauddin Zakariya University Oct 2024 – Mar 2025
- Contributed to research in machine learning and explainable AI by conducting literature reviews, data analysis, and experimental evaluation.
 - Assisted Co-authored research papers on machine learning preprocessing and explainable computer vision models for healthcare applications, currently under review.

PROJECTS

- AI-Powered Healthcare Management System** | Python, Django, Django REST Framework, Scikit-learn, MySQL
- Developed a full-stack healthcare platform featuring patient registration, doctor scheduling, appointment management, electronic medical records (EMR), and secure role-based authentication.
 - Designed RESTful APIs and integrated a Scikit-learn disease risk prediction model to provide real-time clinical decision support while optimizing database queries and backend performance for scalable healthcare operations.
- Brain Tumor Prediction** | Python, Scikit-learn, Machine Learning
- Developed a machine learning model for brain tumor prediction using a tabular clinical dataset, applying data preprocessing, exploratory data analysis (EDA), feature engineering, and feature selection.
 - Performed model training, hyperparameter tuning, and cross-validation while evaluating multiple classification algorithms to identify the best-performing model.
- Customer Churn Prediction** | Python, Scikit-learn, Machine Learning
- Developed a machine learning model to predict customer churn using tabular customer behavioral data. Performed data preprocessing, exploratory data analysis (EDA), feature engineering, and feature selection to prepare high-quality training data.
 - Applied model training, hyperparameter tuning, and cross-validation while comparing multiple classification algorithms using accuracy, precision, recall, F1-score, and ROC-AUC to identify the best-performing model and support customer retention strategies.
- Breast Cancer Prediction** | Python, Scikit-learn, Machine Learning
- Developed a machine learning model for early breast cancer prediction using clinical tabular data. Conducted data preprocessing, exploratory data analysis (EDA), feature engineering, and feature selection to improve data quality and model performance.
 - Trained and evaluated multiple classification algorithms using hyperparameter tuning and cross-validation, selecting the best-performing model based on accuracy, precision, recall, F1-score, and ROC-AUC..

PUBLICATIONS

- Shabbir, A. B. & Amber, K. (2026).** *PRISM: A Framework for Preprocessing Impact Analysis in Machine Learning via Signature-Based Modeling.* UMT Artificial Intelligence Review-Under Review
- Shabbir, A. B. & Amber, K. (2026).** *Explainable Multi-Task Vision Transformer for Automated Autism Spectrum Disorder Detection from Facial Images.* Under Review
- Shabbir, A. B. & Amber, K. (2026).** *Explainable and Fairness-Aware Machine Learning for Early Autism Risk Screening Using Behavioral and Demographic Data.* Under Review

